

Intro to L^AT_EX

A Beginner Guide for Students

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Read this guide next to its source code. Each section shows a command and its output.

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1 Math

L^AT_EX is famous for math. You write math inside math mode. This section shows the basics.

1.1 Math Modes

1.1.1 Inline Math Mode (\$)

Inline math sits inside a sentence. You wrap the math with one dollar sign on each side.

The relationship between three sides of a right angle triangle is $a^2 + b^2 = c^2$.

1.1.2 Display Math Mode (\$\$)

Display math sits on its own line. L^AT_EX centers it. You wrap the math with two dollar signs on each side.

The relationship between three sides of a right angle triangle is

$$a^2 + b^2 = c^2$$

1.2 Superscript

The caret symbol `^` makes a superscript. Braces `{ }` group more than one character.

$$x^2$$

$$x^{31}$$

$$3x^{x^2+5x+2}$$

$$7x^{2^{3^4}}$$

1.3 Subscript

The underscore `_` makes a subscript. Braces group more than one character.

$$a_0$$

$$a_{12}$$

$$a_{0_{12_3}}$$

The command `\ldots` prints clean dots.

$$a_0, a_1, a_2, a_3 \ldots a_{100}$$

1.4 Symbols

1.4.1 Greek Symbols

Greek letters need math mode. A capital first letter gives the capital form.

- α
- β
- γ
- ρ
- σ
- δ
- ϵ
- χ
- π
- Π
- $Area = \pi r^2$

1.4.2 Trigonometry Symbols

L^AT_EX has commands for trig functions. The commands print upright letters.

$$y = \sin(x)$$

$$y = \cos(x)$$

$$y = \tan \theta$$

Some books write the inverse like this.

$$y = \sin^{-1} \theta$$

The command `\arcsin` is clearer. Use it instead.

$$y = \arcsin \theta$$

1.5 Algebraic Symbols

- $\times$
- $\div$
- $\frac{2}{3}$
- $\frac{\sin \theta}{\cos \theta} = \tan \theta$
- $\frac{3x+5}{7x^2+6x+4}$
- $\sqrt{5}$
- $\sqrt[3]{9}$
- $\sqrt[5]{x^2 + y^2}$
- $\sqrt{5 + \sqrt{x^2 + y^2}}$
- $\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{2}}}}}}}$

1.5.1 Other Symbols

These symbols appear a lot in calculus and statistics.

$$\int \quad \oint \quad \Sigma \quad \Pi$$

2 Lists

L^AT_EX has two common list types. The `itemize` environment makes bullet lists. The `enumerate` environment makes numbered lists.

2.1 Bullet Lists with Custom Labels

You can replace a bullet. Put your own label inside square brackets.

- 1 Argentina
 - 2 Switzerland
 - 3 Morocco
 - 4 France
- Spain

- Belgium
- Portugal
- England

2.2 Numbered Lists

The `enumerate` package allows Roman numerals. You can also nest a list inside a list.

- I. Argentina
 - (a) Messi
 - (b) Martinez
 - (c) Lopez
- II. Switzerland
- III. Morocco
- IV. France
- V. Spain
- VI. Belgium
- VII. Portugal
- VIII. England

2.3 More Custom Labels

Empty brackets hide the label. Math symbols also work as labels.

- Argentina
 - (a) Messi
 - (b) Martinez
 - (c) Lopez
- Switzerland
- * Morocco
- * France
- Spain
- Belgium
- = Portugal
- England

3 Tables

The `tabular` environment builds a table. The letters `l`, `c` and `r` set the column alignment. The bar `|` draws a vertical line. The symbol `&` separates cells. The command `\\` ends a row. The command `\hline` draws a horizontal line.

3.1 A Table with a Caption

The `table` environment wraps the `tabular`. It adds a caption and a label. The option `[h]` asks $\text{\LaTeX}$ to place the table here.

Country	Player	Goals
Argentina	Lionel Messi	8
Norway	Erling Haaland	7
France	Kylian Mbappe	7
England	Harry Kane	7

Table 1: Top scorers in the FIFA World Cup 2026 (sample data)

The command `\ref` prints the table number. Table 1 shows sample data.

3.2 A Plain Tabular

A `tabular` without the table wrapper has no caption and no number. A double `\hline` draws a double line.

Country	Player	Goals
Argentina	Lionel Messi	8
Norway	Erling Haaland	7
France	Kylian Mbappe	7
England	Harry Kane	7

4 Text Formatting

4.1 Bold

This is a **bold** text.

4.2 Italics

This is an *italicized* text.

4.3 Nested Styles

This text is ***bold and italicized***.

4.4 Underline

This text is underlined.

4.5 Font Sizes

L^AT_EX has size commands. They run from `\tiny` to `\Huge`.

This is a **huge** word.

This is a **Huge** word.

4.6 Links

The `hyperref` package makes clickable links. The command `\href` takes two parts. The first part is the address. The second part is the visible text.

Today's notes live at [this GitHub page](#).

The command `\url` shows the raw address instead. See <https://github.com/danish-puri/LaTeX>.

5 Graphics

Images need the `graphicx` package. This guide already loads it in the preamble.

5.1 Step 1: Add the Image File

Upload your image to your project folder. Overleaf has an upload button on the top left. Keep the image next to your `.tex` file. Let the image be `img.jpg`.

5.2 Step 2: Insert the Image

The command below inserts the image. Remove the percent sign after you upload `img.jpg`.

The option `width=0.5\textwidth` makes the image half as wide as the text. A fixed size like `width=5cm` also works. The option `scale=0.5` shrinks the image to half size. The option `angle=90` rotates it.

5.3 Step 3: Use the Figure Environment

The `figure` environment adds a caption and a number. It also lets you refer to the image later. Remove the percent signs after you upload the image.

The command `\ref{fig:first}` prints the figure number inside your text.

6 Sectioning and the Table of Contents

L^AT_EX builds structure with sectioning commands. Each command makes a heading at a different level.

- `\section` makes a main heading.
- `\subsection` makes a heading one level down.
- `\subsubsection` goes one more level down.

The command `\tableofcontents` collects every heading. It prints them at the start with page numbers. Compile the document twice. The first run collects the headings. The second run prints the table.

The next two sections show this structure in action. Find them in the table of contents at the start of this guide.

7 Polynomials

A polynomial is a sum of powers of x . The degree is the highest power.

7.1 Linear

A linear polynomial has degree one. Its graph is a straight line.

$$y = mx + c$$

Here m is the slope and c is the intercept.

7.2 Quadratic

A quadratic polynomial has degree two. Its graph is a parabola.

$$y = ax^2 + bx + c$$

The quadratic formula gives the roots.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

7.3 Cubic

A cubic polynomial has degree three.

$$y = ax^3 + bx^2 + cx + d$$

8 Calculus

8.1 Limits

The command `\lim` prints the limit symbol. The arrow uses `\to`.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

8.2 Derivative

The derivative measures the rate of change.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The power rule is the most common rule.

$$\frac{d}{dx} x^n = nx^{n-1}$$

8.3 Integration

The command `\int` prints the integral sign. The limits use subscript and superscript.

$$\int x^2 dx = \frac{x^3}{3} + C$$
$$\int_0^1 x^2 dx = \frac{1}{3}$$

8.4 Differential Equations

A differential equation contains a function and its derivatives.

8.4.1 First Order Differential Equations

A first order equation contains the first derivative only.

$$\frac{dy}{dx} + P(x)y = Q(x)$$

8.4.2 Second Order Differential Equations

A second order equation contains the second derivative.

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + c y = 0$$

9 Where to Go Next

Practice is the best teacher. Rebuild your last assignment in $\text{\LaTeX}$. Use this guide as your reference.

These free resources will help you go further.

- Overleaf documentation: <https://www.overleaf.com/learn>